Agent Fidelity Spectrum for Government AI

A five-level framework for measuring AI agent response fidelity

across Copilot Studio, Foundry, and pro-code architectures

Whitepaper | v2.0

March 2026

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L1: Discovery	L2: Summarization	L3: Operational	L4: Investigative	L5: Adjudicative

Not all AI use cases require the same level of response fidelity, and not all agent architectures deliver it. This report presents a five-level framework based on 757 empirical test runs across 34 agent configurations and three government use cases.

757 test runs | 34 agents | 6 models | 45 prompts | 3 use cases

The agent accelerates the human; it does not replace them. Mandatory human review, citation linking, audit logging, and organizational culture that treats AI output as a draft, never a decision. Aligned with the NIST AI Risk Management Framework. This is the only responsible operating model.

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At a Glance

Summary

This framework is grounded in 757 test runs across 34 agent configurations and 6 models, 3 government use cases, and multiple testing rounds. Four gaps emerged: model, data, tool, and retrieval. Every gap was fixable, and none required AI expertise. Better tools, cleaner data, cross-referenced documents, and the right retrieval infrastructure. Here is what the data says.

Levels 1 - 2

Just Works

8/10 with zero customization. Model choice irrelevant.

Level 3

Data Over Documents

MCP outperformed document search on every aggregate query.

Level 4

The Inflection Point

Four gaps emerged: model, data, tool, and retrieval. All fixable.

Level 5

Human in the Loop

High fidelity at L4 means humans review conclusions, not raw data.

The Five Levels

- **L1** Discovery
- **L2** Summarization
- **L3** Operational
- **L4** Investigative
- **L5** Adjudicative

Three Use Cases

Meet Your Agents

M365 Copilot
Zero code, platform model

Copilot Studio
Highly configurable

Foundry
Agent Service, managed

Pro-code
Full control, custom SDK

Full agent reference in Appendix.

Use Case 1

Legal Case Analysis

A child welfare legal office prepares for hearings by reconstructing timelines, identifying conflicting witness statements, and flagging discrepancies across case files.

50 cases **277** people **333** events **151** discrepancies

17 agent configurations

Use Case 2

Investigative Analytics

A municipal investigative unit cross-references property ownership, code violations, and business registrations to identify patterns of landlord negligence.

34M rows **584K** properties **1.6M** violations

8 agent configurations

Use Case 3

Policy Compliance

A state Medicaid agency tests chatbot accuracy for caseworker policy lookups across eligibility rules, resource limits, and procedural guidance.

37 chapters **25** questions **12** agents

12 agent configurations

How This Was Built

This project - the code, the data, and the documentation you are reading - was AI-accelerated. One solution engineer. One AI coding assistant. Three agency engagements. Six weeks.

Human-directed. Architecture decisions, Azure service selection, schema design, security posture, and the five-level fidelity framework came from domain expertise and customer conversations.

AI-accelerated. TypeScript Functions, MCP server, SQL schema, 50 synthetic cases (277 people, 333 timeline events), web UI, eight PDF generators, a 26-slide deck, and 37 enriched policy documents were generated by AI and reviewed by a human.

Human-verified. 757 test runs across 34 agent configurations and 6 models, scored by hand. Every dangerous response - hallucinated facts, missed fractures, invented timelines, fabricated policy procedures - was caught through manual review. That is exactly the point of Level 5.

The Five Levels

Government agencies deploy AI agents across a spectrum of use cases. Each level has different fidelity requirements, different failure consequences, and different engineering investments. This framework helps leaders match their investment to their risk.

Level	Name	Stakes	Example
1	Discovery	Minor inconvenience	Find our leave policy
2	Summarization	Wasted time	Summarize this audit report
3	Operational	Misallocated resources	How many open cases by type?
4	Investigative	Missed evidence	Build a timeline from these records
5	Adjudicative	Wrong legal outcome	Prepare facts for this hearing

1 Discovery

2 Summarization and Synthesis

The simplest AI use cases: finding documents, summarizing reports, condensing meeting transcripts. Wrong answers are inconvenient but not dangerous. Users can easily verify results against the original source.

What we found: Document-based agents (Copilot Studio with SharePoint grounding) scored 8 out of 10 on both use cases with zero customization. **GPT-4o (Government Cloud) and GPT-4.1 (Commercial) performed identically at this level.** Model choice had no measurable impact at these levels. Invest in document hygiene (cross-reference headers that point reviewers to related documents, consistent formatting, meaningful filenames, metadata tags) rather than custom engineering.

Use Case 3 confirmed: Use Case 3 reinforced this dramatically: the same enriched Medicaid policy documents scored 39% through SharePoint search but 91% through knowledge base upload. Document quality is necessary but not sufficient - **retrieval infrastructure determines whether good documents produce good answers.** See the Use Case 3 section for the full retrieval gap analysis.

Recommendation: Deploy a Copilot Studio agent with SharePoint or a knowledge base. This is where most agencies should start. Test retrieval accuracy before investing in custom engineering.

3 Operational Decision Support

Multi-source data aggregation, dashboards-via-conversation, workload triage. Wrong answers misallocate resources but are verifiable against existing reports.

What we found: Aggregate queries (portfolio summaries, citywide statistics, workload counts) worked reliably across all agent types, including the weakest performers. **This is where agents with structured data access begin to outperform document agents,** including both Copilot Studio and pro-code agents. Live structured data delivers

answers that static documents cannot.

Recommendation: Connect agents to structured data sources via Model Context Protocol. If your operational data already lives in Dataverse, the built-in Dataverse MCP connector provides this path with zero custom code. Write clear tool descriptions. Add summary modes for large result sets.

4 Investigative and Analytical

Timeline reconstruction, discrepancy detection, entity resolution across multiple data sources. Wrong answers mean missed evidence, false leads, and misdirected investigations. This is where response fidelity diverges dramatically.

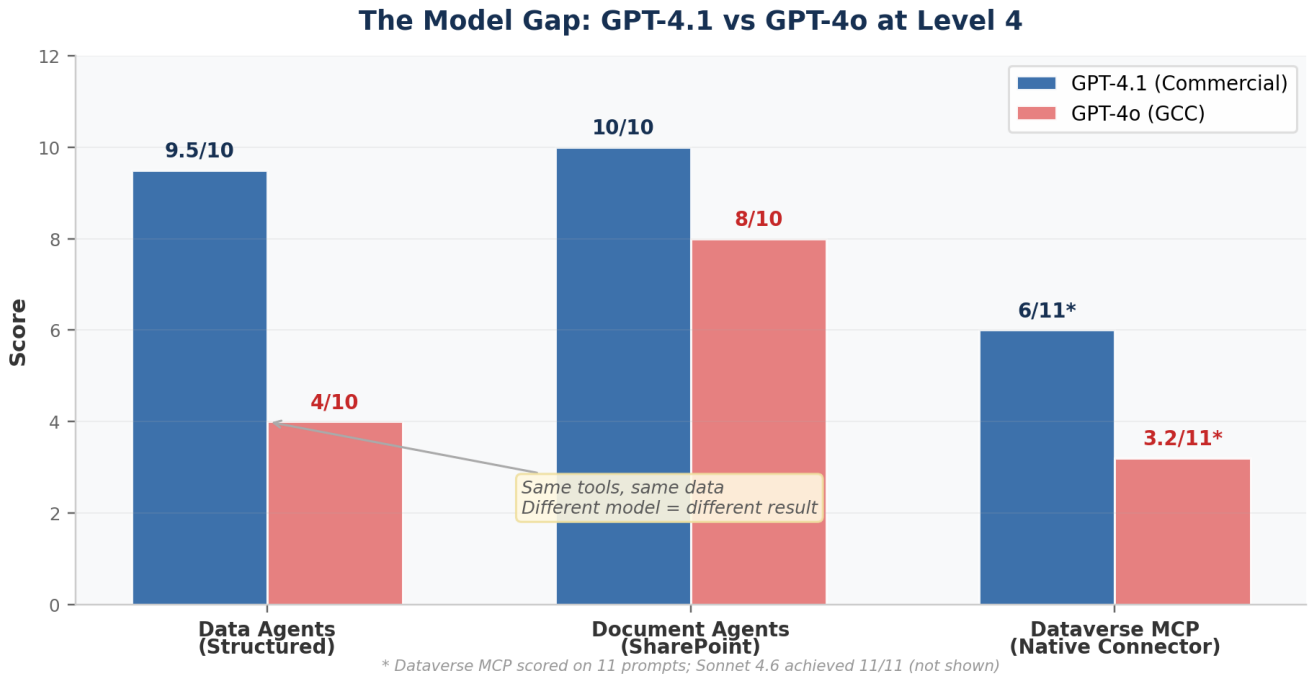
The Model Gap

Among agents with structured data access, GPT-4.1 agents averaged 9.5 out of 10. The GPT-4o agent scored 4 out of 10. Same tools, same data, same backend. Tool and prompt engineering alone did not close this gap at Level 4.Document agents showed no model gap (8 out of 10 on both GPT-4o and GPT-4.1).

Model	Average Score	Platform
GPT-4.1	9.5 / 10	Commercial Copilot Studio, Pro-code agents
GPT-5 Auto	9.5 / 10	Commercial Copilot Studio (identical to 4.1)
GPT-4o	4 / 10	Government Cloud Copilot Studio
Platform-assigned	2 / 10	M365 Copilot Commercial (no user control)

** Ranked by score*

Dataverse MCP multi-model test (5 models): GPT-4o 3.2/11, GPT-5 Auto 4/11, GPT-4.1 6/11, GPT-5 Reasoning 10/11, Sonnet 4.6 11/11.



The Data Gap

The agent correctly queried the database, found no results, and concluded no drug tests existed. The answer was wrong because the data was incomplete, not because the agent failed.

Issue	In Source Documents	In Database	Agent Result

Fentanyl screens	Yes (narrative text)	Not extracted	"No drug tests exist"
Skeletal survey	Yes (medical records)	Not extracted	Could not answer prompt

Recommendation: Audit your data before auditing your agent. If critical facts live only in unstructured reports, they must be extracted and loaded before an agent can query them.

The Tool Gap

Users type '4763 Griscom Street.' The database stores '4763 GRISCOM ST.' Without a tool to bridge that gap, agents resolved the wrong property 87% of the time. Every downstream answer was confidently wrong. One address normalization tool fixed it with simple engineering.

Issue	User Input	In Database	Agent Result
Street address	"4763 Griscom Street"	"4763 GRISCOM ST"	Wrong property 87% of the time

Recommendation: Purpose-built tools for entity resolution. Models with strong multi-step reasoning (GPT-4.1 in our testing).Explicit workflow guidance in system prompts. Ground truth test suites with known answers.

The Retrieval Gap

Use Case 3 revealed a fourth gap: even when documents contain the right answer, the retrieval infrastructure determines whether the agent finds it. The same enriched Medicaid policy documents produced dramatically different accuracy depending on how they were indexed and searched.

Retrieval Method	Accuracy	Notes
SharePoint search (baseline)	39%	Search index surfaced wrong chapters consistently
SharePoint + cross-references	52%	Document enrichment improved but did not solve retrieval
SharePoint + cross-refs + metadata	61%	Six custom metadata columns boosted search relevance
Knowledge base upload + cross-refs	91%	Bypassed SharePoint search entirely; top performer
Azure AI Search + enriched docs	91%	Purpose-built retrieval; tied KB upload
CPS + AI Search (production agent)	91%	Copilot Studio front end with AI Search backend

Recommendation: Before investing in model upgrades or prompt engineering, verify that your retrieval infrastructure surfaces the right documents. If it does not, no model will produce the right answer. Knowledge base upload and Azure AI Search both solved the retrieval problem in our testing.

5 Legal and Adjudicative

Case preparation, evidence compilation, hearing summaries, regulatory determinations. Wrong answers can produce incorrect legal outcomes, due process violations, or direct harm to vulnerable populations.

What we found: Even our highest-scoring agents produced dangerous results. Each example below was documented during testing:

Failure Type	What Happened	Legal Risk
Reproduced misinformation	Quoted 'no fractures detected' from sheriff report; medical records showed bilateral long bone fractures	Wrong medical conclusion in child abuse investigation
False negative	Found two positive fentanyl screens, then concluded no drug tests existed	Missed evidence contradicting sworn testimony
Misattribution	Attributed mother's '8 PM bedtime' statement to father (who consistently said 'around ten')	Undermined witness credibility at hearing
Fabricated procedure	Invented a 4-step COLA disregard process with specific dollar amounts not in the policy manual	Caseworker applies non-existent policy to eligibility decision

Recommendation: Mandatory human review by a qualified professional before any legal action. Citation linking so reviewers can verify every claim. Audit logging for accountability. Treat every agent output as a draft, never a decision.

The Iterative Process

Deploying an AI agent is not a one-time event. Each round of testing reveals a different category of failure requiring a different type of fix. Organizations that skip this process will encounter every failure mode documented here. This section covers Use Cases 1 and 2. Use Case 3 follows in a dedicated section.

Round 0: Test Everything

Without ground truth, you cannot measure improvement. We tested 21 agent configurations across 20 prompts before making any changes.

Document Agents

12 document agent configurations tested across 10 prompts each (120 test runs). Scores ranged from 3 out of 10 to 8 out of 10. Seven of eight missed the skeletal survey discrepancy because the platform retrieved only one of two conflicting documents.

Data Agents

7 structured data agent configurations tested across 10 prompts each (70 test runs). Scores ranged from 0 out of 10 to 8 out of 10. The best agent still missed 2 prompts due to missing data. Address resolution failed 87% of the time with no tool to bridge user input to database identifiers.

What We Found

Testing revealed five categories of AI failure, ranked by severity. These are not hypothetical. Each was documented across 462 test runs. **Every failure below occurred in agents that ultimately scored 9 or 10 out of 10** - which is why human review remains essential regardless of final scores.

Severity	Failure Mode	Description	Example	Agent(s)
Critical	**False negative**	Data retrieved but not recognized as the answer	Found fentanyl screens, then concluded no drug test results existed	Data Agent
Critical	**Reproduced misinformation**	Quoted a source whose characterization was misleading in isolation	Sheriff: 'no fractures'; medical records showed bilateral fractures	Document Agent
High	**Misattribution**	Correct fact assigned to the wrong person	Mother's '8 PM' attributed to father (who said 'around ten')	Document Agent
High	**Hallucinated fact**	Invented specific detail with no source	ER arrival at '2:00 AM'; actual time was 3:15 AM	Data Agent
Medium	**Silent failure**	No results returned without warning	Hallucinated case number; returned 1 of 9 results	Both

Baseline Scores

Agent	Score

Case Analyst Agent (pro-code / MCP Chat)	10/10
Copilot Studio MCP (Commercial)	8/10
Copilot Studio SharePoint/PDF (Commercial)	8/10
Foundry Agent (Foundry Agent Service)	4/10
Copilot Studio SharePoint/PDF (Gov Cloud)	3/10
Copilot Studio MCP (Gov Cloud)	2/10
Investigative Agent (pro-code / OpenAI SDK)	1/10
Copilot Studio Dataverse MCP (Gov Cloud)	1/10
Triage Agent (pro-code / Semantic Kernel)	0/10

** Ranked by score. Data agents scored on Use Case 2 (investigative analytics). Document agents scored on Use Case 1 (legal case analysis). Both document agents also scored 8/10 on UC2 (stable across all rounds).*

Round 1: Fix the Foundations

Round 1 targeted two categories: missing data in the database and missing cross-references in documents.

Data Agents

What changed: Narrative facts (drug screens, skeletal survey) were not in the database. A new address normalization tool eliminated the 87% address resolution failure rate. Fix: 11 new SQL rows, 1 tool filter, 1 new tool.

Document Agents

What changed: Cross-reference headers added to all documents listing related documents and their key findings. No content was altered. Zero code.

Round 1 Scores

Agent	Round 0	Round 1
Case Analyst Agent (pro-code / MCP Chat)	10/10	10/10
Copilot Studio MCP (Commercial)	8/10	10/10
Copilot Studio SharePoint/PDF (Commercial)	8/10	10/10
Investigative Agent (pro-code / OpenAI SDK)	1/10	8/10
Foundry Agent (Foundry Agent Service)	4/10	8/10
Copilot Studio SharePoint/PDF (Gov Cloud)	3/10	7/10
Copilot Studio MCP (Gov Cloud)	2/10	4/10
Copilot Studio Dataverse MCP (Gov Cloud)	1/10	1/10
Triage Agent (pro-code / Semantic Kernel)	0/10	1/10
M365 Copilot MCP (Commercial)	--	2/10

** Ranked by Round 1 score. M365 Copilot first tested in Round 1. UC2 scores for data agents; UC1 scores for document agents.*

Why M365 Copilot scored lowest: M365 Copilot scored 2/10. Its platform-assigned model, limited tool customization, and lack of system prompt control made iterative improvement impractical. Organizations with existing

M365 Copilot licenses should use it for Level 1 tasks (document discovery) and deploy Copilot Studio for anything higher.

Round 2: Refine and Optimize

Round 2 refined MCP tool descriptions, agent system prompts, and SharePoint metadata tags.

Data Agents

What changed: Triage Agent underwent five sub-rounds of system prompt and tool description refinement, climbing from 1/10 to 10/10. Investigative Agent reached 10/10 with minor tool description adjustments. Foundry Agent improved to 9/10 but its managed orchestration layer limited tool sequencing on the hardest prompt. GCC MCP remained at 4/10 - the same improvements that lifted other agents had no effect with GPT-4o at this level, confirming a model-level ceiling for Level 4 workloads.

Document Agents

What changed: SharePoint metadata tags (document topics, keywords) boosted retrieval for queries the platform previously missed. Cross-reference headers on a second case file fixed a missing-document retrieval gap. Still zero code.

Additional agents retested: All other document agents also benefited from cross-referenced documents: KB/DOCX/Com improved from 6/10 to a perfect 10/10, SP/DOCX/Com from 7/10 to 9/10, KB/PDF/GCC from 5/10 to 9/10, KB/PDF/Com from 5/10 to 8/10, KB/DOCX/GCC from 4/10 to 8/10, and SP/DOCX/GCC from 5/10 to 7/10. The same document hygiene improvements delivered gains across upload method, file format, and cloud environment.

Round 2 Scores

Agent	Round 0	Round 1	Round 2
Case Analyst Agent (pro-code / MCP Chat)	10/10	10/10	10/10
Copilot Studio MCP (Commercial)	8/10	10/10	10/10
Investigative Agent (pro-code / OpenAI SDK)	1/10	8/10	10/10
Triage Agent (pro-code / Semantic Kernel)	0/10	1/10	10/10
Copilot Studio SharePoint/PDF (Commercial)	8/10	10/10	10/10
Copilot Studio DV MCP / Sonnet 4.6 (Commercial)	--	--	10/10
Copilot Studio DV MCP / GPT-5 Reasoning (Commercial)	--	--	10/10
Copilot Studio SharePoint/PDF (Gov Cloud)	3/10	7/10	9/10
Foundry Agent (Foundry Agent Service)	4/10	8/10	9/10
Copilot Studio Dataverse MCP / GPT-4.1 (Commercial)	--	--	6/10
Copilot Studio MCP (Gov Cloud)	2/10	4/10	4/10
Copilot Studio DV MCP / GPT-5 Auto (Commercial)	--	--	4/10
Copilot Studio Dataverse MCP (Gov Cloud)	1/10	1/10	3.2/10
M365 Copilot MCP (Commercial)	--	2/10	--

** Ranked by Round 2 score. M365 Copilot not retested (platform constraints prevent iterative improvement). Dataverse MCP agents scored on UC1 prompts against structured Da*

Combined result: Six agents reached 10/10 (including KB/DOCX/Com, which jumped from 6/10 to a perfect score). Copilot Studio SP/PDF/GCC improved from 3/10 to 9/10 with zero code changes. Every retested document agent improved. The Triage Agent climbed from 0/10 to 10/10 across five sub-rounds of prompt and tool refinement. Copilot Studio MCP/GCC plateaued at 4/10 despite receiving every improvement the other agents received, confirming a model-level ceiling for these workloads with GPT-4o. A multi-model test of the native Dataverse MCP connector proved the model is everything: GPT-4o scored 3.2/11 (best after 6 rounds of optimization), GPT-5 Auto 4/11, GPT-4.1 6/11, GPT-5 Reasoning 10/11, and Sonnet 4.6 achieved a perfect 11/11 on identical schema and data across 5 models. GPT-5 Reasoning nearly matches Sonnet, losing only on a connector-level abbreviation bug. GPT-5 Auto still performed worse than GPT-4.1.

The Visual Story

The heatmap below shows how scores progressed across testing rounds. Green cells represent high fidelity; red cells represent low fidelity. Every agent above the diagonal improved. The pattern is clear: iterative testing and targeted fixes work.

Score Progression Across Testing Rounds (UC1 & UC2)

Iterative Improvement

	Round 0 Baseline	Round 1 Fix Foundations	Round 2 Refine & Optimize
Case Analyst (MCP)	10/10	10/10	10/10
CS MCP/Com	8/10	10/10	10/10
CS SP/PDF/Com	8/10	10/10	10/10
Investigative (SDK)	1/10	8/10	10/10
Triage (SK)	0/10	1/10	10/10
Foundry Agent	4/10	8/10	9/10
CS SP/PDF/GCC	3/10	7/10	9/10
CS MCP/GCC	2/10	4/10	4/10
CS DV MCP/GCC	1/10	1/10	3.2/10

The Improvement Playbook

Every organization that deploys AI agents should follow this sequence. Skipping steps does not save time - it hides failures until production.

1. **Test with known answers**
   Without ground truth, you cannot measure improvement. Build a test suite before you build the agent.
2. **Fix the data**
   If the answer is not in the data, no model will find it. Make facts discrete and queryable.
3. **Fix the tools**
   If the model cannot reach the data, add tools. If it does not know which tool to use, improve descriptions.
4. **Fix the documents**
   Invest in document hygiene before custom engineering. Cross-reference headers, metadata tags, and consistent formatting compound across every document agent.

5 **Fix the retrieval**

Verify that the right documents are being surfaced. If SharePoint search is the bottleneck, test knowledge base upload or Azure AI Search as alternative retrieval paths.

6 **Retest after every change**

Each fix can introduce new failure modes. Run the full test suite after every change.

Phase 2: Structured API Without MCP

Phase 1 proved that MCP-backed agents with structured data access dramatically outperform document-grounded agents at Level 4. But MCP requires a Container App, custom TypeScript code, and APIM configuration. Many government teams want the fidelity gains without the infrastructure investment.

Phase 2 tests whether a Power Platform custom connector - which requires zero code and deploys in under 30 seconds - can deliver comparable fidelity by giving Copilot Studio the same structured API access through a different path.

Phase 2 Hypothesis

GCC customers can improve Copilot Studio agent fidelity by adding structured API tools via custom connectors, without needing MCP infrastructure. A Swagger file and a single CLI command is all it takes.

What We Built

The custom connector creates a new data path that reuses the existing backend:

Copilot Studio Agent -> Custom Connector -> APIM -> Azure Functions -> Azure SQL

No new backend code was written. The connector calls the same 5 APIM operations the MCP server uses. Any fidelity difference is attributable to the orchestration layer, not the data. The entire process took four steps:

1. **Export the APIM Swagger spec**
   APIM already had a Swagger 2.0 definition. One Azure CLI command exported it.
2. **Enrich the Swagger spec**
   Added detailed operation descriptions, parameter metadata, example values, and full response schemas. Copilot Studio uses this metadata to decide when and how to call each tool.
3. **Generate the API properties file**
   Power Platform requires a properties file alongside the Swagger spec. One pac CLI command generated the template.
4. **Deploy to GCC**
   A single command: pac connector create. The connector appeared in the GCC environment in under 30 seconds. All 5 operations were immediately available as tools in Copilot Studio.

Operation	Route	Description
ListCases	GET /dss/cases	All cases with metadata
GetCaseSummary	GET /dss/cases/{caseId}	Case detail + people involved
GetTimeline	GET /dss/cases/{id}/timeline	Events, optional type filter

GetStatements	GET /dss/cases/{id}/statements	Witness statements, person filter
GetDiscrepancies	GET /dss/cases/{id}/discrepancies	Contradictions between accounts

Results: 8.5 / 10

We ran all 19 Phase 1 evaluation prompts against the custom connector agent. The zero-code agent scored 161/190 (8.5/10), achieving 85% of MCP fidelity with no custom infrastructure.

Section	Score	Notes
Factual Retrieval (4)	10.0/10	All 4 prompts perfect. 14 timeline events, all 8 people, all hearings.
Cross-Referencing (3)	9.3/10	Exact quotes, side-by-side comparisons. Minor: missed one analytical shift.
Discrepancies (3)	6.7/10	Two perfect. One failure: could not resolve ambiguous case reference.
Filtering (3)	7.0/10	Two perfect. One failure: resolved 'Webb case' to wrong Webb case.
Aggregate (4)	9.3/10	Counted 25 active cases, listed all TPR cases (9). Minor omission on one.
Stress Tests (2)	7.5/10	Correct references but missed cross-references and net calculations.

Failure Analysis

The 2 hard failures (0/10 and 1/10) were both orchestration issues, not data access issues. The connector delivered data correctly on every other prompt. The failures occurred when GPT-4o could not resolve ambiguous case references:

- **Prompt 3.3 (0/10):** Says 'Jaylen' and 'the parents' with no case ID. GPT-4o did not attempt a ListCases lookup to resolve the implicit reference. The MCP agent handles this because its orchestration loop is more aggressive about disambiguation.
- **Prompt 4.1 (1/10):** Says 'the Webb case' but there are two Webb cases in the database. GPT-4o resolved to the wrong one. When the prompt includes an explicit case ID, the connector agent performs at MCP-level fidelity.

Pattern: When prompts include explicit identifiers (case ID, full names), the custom connector agent scores 10/10 consistently. The gap appears only when the agent must reason about which entity the user means - an orchestration problem, not a data access problem. System prompt instructions telling the agent to always resolve case IDs via ListCases before calling other tools may close this gap entirely.

Custom Connector vs. MCP

Dimension	MCP Server	Custom Connector
Code required	TypeScript MCP server	None (Swagger file only)
Infrastructure	Container App + ACR	None
Deployment	Docker build + deploy	pac connector create
Tool discovery	MCP protocol	Swagger operation IDs
Orchestration	Server-side (MCP loop)	Copilot Studio (platform)

Model control	Choose any model	Platform-assigned (GPT-4o GCC)
GCC available	Yes (custom infra)	Yes (native)
Fidelity (UC1)	10/10	8.5/10

The Takeaway

A zero-code custom connector delivers 85% of MCP fidelity using the same data, same endpoints, and the same evaluation prompts. For many government use cases, this is enough. For the hardest prompts requiring multi-step disambiguation, MCP's server-side orchestration or a stronger model still wins.

Use Case 3: Policy Compliance

A state Medicaid agency uses a chatbot to help caseworkers look up eligibility rules, resource limits, and procedural guidance from the Medicaid Policy and Procedures Manual (MPPM). The existing chatbot had been deployed for several months. The agency wanted to know: how accurate is it, and can we make it better?

We tested 12 agent configurations against 25 test questions developed by policy experts. Questions ranged from simple lookups ('What is the burial fund resource limit for a couple?') to nuanced policy interpretation ('If a member applying for ABD reports life insurance with face value $5,000 and cash surrender value $3,000, is that resource counted under the liberalized policy?').

Auditing the Baseline

The existing chatbot (GCC Knowledge Base with original documents) was graded at 70% accuracy (16/23) by the customer. When we independently reviewed each response against the actual policy text, three answers marked 'correct' were factually wrong. The adjusted accuracy was 57% (13/23).

Question	Customer Grade	Independent Review	Issue
COLA disregard	Correct	Incorrect	Fabricated a 4-step procedure with invented dollar amounts not in the policy
E-signatures on Form 1282	Correct	Incorrect	Stated documents do not mention e-signatures; policy explicitly allows them
Life insurance for ABD	Correct	Incorrect	Applied $1,500 SSI threshold instead of $10,000 liberalized ABD threshold

Key takeaway: Five additional responses marked correct or left ungraded contained significant omissions or inaccuracies. Self-reported accuracy is not ground truth. Independent review against source policy documents is essential.

The Retrieval Problem

The primary driver of incorrect answers was document retrieval failure. The chatbot consistently retrieved the wrong policy chapters and based its answers on irrelevant content. Section 300 (Non-MAGI resource and income policy) was the largest blind spot: 7 of 11 failures involved questions that expected Section 300 content, but the chatbot retrieved Chapter 403 (Optional State Supplementation), Chapter 601 (Presumptive Eligibility), or Chapter 703 (Disability Determination) instead.

The content existed in the enriched documents. The search index failed to surface it.

What We Changed

Step 1: Document Enrichment

The original 20 source documents were processed through an automated enrichment pipeline that produced 37 individual chapter files. Each enriched document included metadata headers, inline cross-reference expansions, and

a cross-reference guide listing all related sections. Result: SharePoint accuracy improved from 39% to 52%.

Step 2: SharePoint Metadata

Six custom metadata columns were added to the SharePoint document library: Program Type, Key Thresholds, FAQ (natural-language questions), MPPM Section, and Chapter Number. Result: SharePoint accuracy improved from 52% to 61%.

Step 3: Alternative Retrieval

The enriched documents were tested through three alternative retrieval paths: knowledge base upload (bypassing SharePoint search entirely), Azure AI Search (purpose-built retrieval infrastructure), and Copilot Studio connected to Azure AI Search via Power Automate. All three achieved 91% accuracy (21/23) - a 52 percentage point improvement over the SharePoint baseline with the same documents.

Content Gaps

Three questions remained unanswered across all 12 agents because the source policy documents do not contain the answer. No amount of engineering can fix missing source content.

Gap	Issue	Recommendation
1099-R verification	Only 1 mention in entire 37-document corpus. No guidance on pension income verification.	Add 1099-R section to income policy chapters
Veterans/VA in MAGI context	35 references in Non-MAGI chapters, 0 in MAGI chapters. Chatbot cannot find relevant policy.	Add VA benefits cross-reference to MAGI income chapter
Deemed eligible newborn	Only covers adoption scenarios, not general newborn deeming rule.	Expand eligibility chapter to cover all newborn deeming scenarios

Use Case 3 Scores

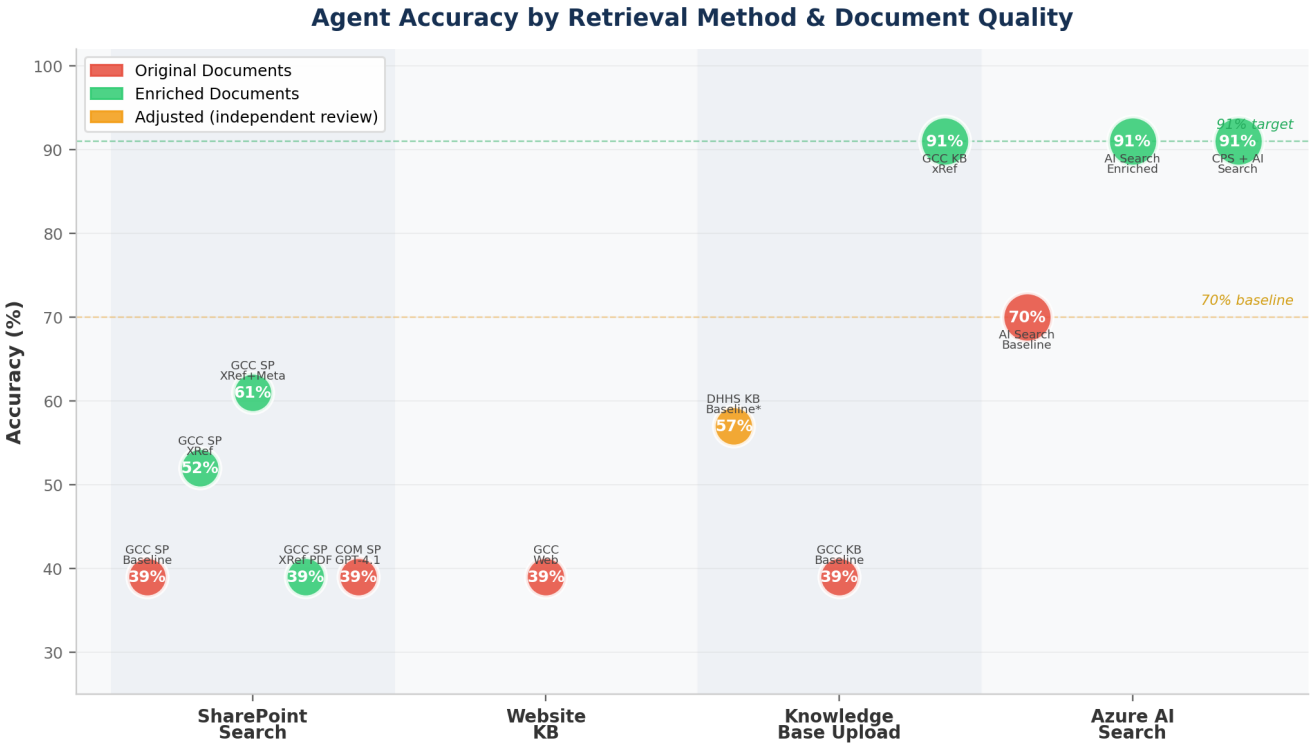
Agent Configuration	Score	Accuracy
GCC KB + Cross-Referenced Docs	21/23	91%
Azure AI Search + Enriched Docs	21/23	91%
CPS + AI Search + Enriched Docs (production agent)	21/23	91%
Azure AI Search + Original Docs	16/23	70%
GCC SP + Cross-Refs + Metadata	14/23	61%
GCC SP + Cross-Referenced Docs	12/23	52%
DHHS GCC KB Baseline (customer-reported)	16/23	70%
DHHS GCC KB Baseline (adjusted)	13/23	57%
GCC SP Baseline	9/23	39%
Commercial SP Baseline (GPT-4.1)	9/23	39%
GCC Web (Direct Line API)	9/23	39%

GCC KB + Original Docs	9/23	39%
GCC SP + Cross-Refs as PDF	9/23	39%

** Ranked by accuracy. Questions 12 and 16 excluded from scoring (not tested across all agents). SP XRef PDF = 39% confirms SharePoint search is the bottleneck, not file format. Commercial SP Baseline (GPT-4.1) = 39% confirms the model does not matter when retrieval fails.*

The Retrieval Gap, Visualized

Same enriched documents, four retrieval methods, dramatically different results. Bubble size reflects accuracy threshold. Green = enriched documents, red = original documents, orange = adjusted after independent review.



The Retrieval Gap in One Line

SharePoint search: 39-61%. Knowledge base upload: 91%. Azure AI Search: 91%. The retrieval infrastructure is as important as document quality.

Government Cloud (GCC) Options

The Model Gap

Government Community Cloud currently defaults to GPT-4o for Copilot Studio. For agents with structured data access, this model scored 4 out of 10 on Level 4 investigative queries where GPT-4.1 scored 9.5 out of 10. Document agents were unaffected (8 out of 10 on both models). Tool and prompt engineering alone did not close the structured data gap at Level 4.

Organizations on Government Cloud have seven practical paths forward today:

1. **Improve document quality**
   Cross-reference headers and SharePoint metadata tags improved the Government Cloud document agent from 3 out of 10 to 9 out of 10, surpassing the Commercial agent. Zero code, zero model dependency.
2. **Fix the retrieval path**
   Use Case 3 proved that GCC Knowledge Base upload with enriched documents achieved 91% accuracy where SharePoint search peaked at 61%. If SharePoint search is the bottleneck, switch to KB upload or Azure AI Search.
3. **Use Copilot Studio document agents for Levels 1-3**
   GPT-4o is adequate for retrieval and summarization.
4. **Deploy a Foundry Agent**
   Zero to low code (portal or SDK) using Azure OpenAI GPT-4.1 for Level 4 and 5 workloads.
5. **Build a fully custom agent**
   Use Azure OpenAI GPT-4.1 directly for maximum control over orchestration, tools, and evaluation.
6. **Use the Dataverse MCP connector**
   If case data lives in Dataverse, give Copilot Studio agents structured data access without building custom APIs.
7. **Monitor GCC model updates**
   When GPT-4.1 becomes available in Government Cloud, Copilot Studio agents will benefit immediately.

These options are not mutually exclusive. Most organizations will combine document agents for everyday queries with a structured data agent for investigative workloads. The table below shows how engineering investment scales across these approaches.

The Code Investment Spectrum

This table shows the agent-side investment. MCP-connected agents need an MCP server. If your data lives in Dataverse, the built-in Dataverse MCP connector eliminates that step. For any other data source, building a custom MCP server and backing APIs is a separate pro-code investment. Once that foundation is in place, the agent-side engineering changes what you can customize, not whether the agent works.

Approach	Code Required	Best For
M365 Copilot + MCP (Com)	Config only	Levels 1-2
Copilot Studio + SharePoint	Config only	Levels 1-3
Copilot Studio + KB Upload	Config only	Levels 1-3 (improved retrieval)
Copilot Studio + Custom Connector	Config only	Levels 1-4 (8.5/10, zero code)
Copilot Studio + MCP	Config only	Levels 1-4 (Com), 1-3 (GCC)
CPS + AI Search + Power Automate	Low	Levels 1-3 (best retrieval)
Foundry Agent	Low	Levels 3-4
OpenAI SDK / Semantic Kernel	Full	Levels 4-5

Every architecture scored 9 to 10 out of 10 with GPT-4.1. The investment buys customization, governance controls, and audit capabilities, all of which matter most at Levels 4 and 5.

Why Copilot Studio

Copilot Studio is the platform that ties the spectrum together. It hosts both zero-code SharePoint agents (Levels 1-2) and MCP-connected structured data agents (Levels 3-4) under a single governance boundary. For agencies with data in Dataverse, the built-in MCP connector means zero custom code for structured data access. Model flexibility is the headline: the same Copilot Studio agent configuration scored 4 out of 10 with GPT-4o and 10 out of 10 with GPT-4.1 with no configuration changes required. **When Government Cloud gains access to models with stronger multi-step reasoning, every Copilot Studio agent improves overnight.** Enterprise governance (data loss prevention, audit logging, admin pre-approval for tool calls) and M365 distribution (agents appear in Teams and Copilot chat) are built into the platform, not bolted on after the fact.

Six Findings That Surprised Us

1. The most dangerous agent was the most faithful to its source

Seven of eight document agents faithfully reproduced a sheriff's report statement about fracture findings while the medical records told a different story. The agents were not wrong about what the document said. They were wrong about what was true.

2. A single missing tool caused an 87% failure rate, and simple engineering fixed it

No tool existed to convert a street address into a database identifier. Adding one address lookup tool produced zero failures in retesting. One agent went from 1 out of 10 to a perfect 10 out of 10. The cost of not having the tool was enormous; the cost of building it was trivial.

3. The most complex agent needed five rounds to reach perfection

The Semantic Kernel team-of-agents pattern started at 0 out of 10 and reached a perfect 10 out of 10, but only after five rounds of sub-agent prompt engineering. A simpler Copilot Studio agent scored 8 out of 10 with no customization at all.

4. The model retrieved the answer and did not recognize it

The agent called the right tools, received data containing 'two positive drug screens (fentanyl) in October,' and concluded: 'no drug test results exist.' This false negative is invisible to users because the tool calls look correct.

5. An advanced agent challenged its own premise, and was right

The Foundry Agent was asked about a purchase price versus assessment ratio. Instead of calculating the answer, it checked the source data, found the premise was wrong, and recalculated using verified numbers. No other agent questioned the input.

6. The customer's own accuracy grading was inflated by 13 percentage points

When we independently reviewed the existing Medicaid policy chatbot's responses against the actual policy documents, three answers marked 'correct' by the customer were factually wrong - fabricated procedures, wrong thresholds, and misapplied policy frameworks. The true baseline was 57%, not 70%. You cannot improve what you cannot accurately measure, and self-reported accuracy is not reliable ground truth.

Risk Management and Responsible AI

The fidelity framework presented in this report is not just a testing methodology - it is a risk-tiering system. Each level maps increasing consequence of failure to increasing investment in tools, data, models, and human oversight. This aligns directly with NIST's AI Risk Management Framework (AI 100-1), which provides the most widely recognized standard for managing AI risk in government.

Mapping to NIST AI RMF

The four core functions of NIST AI RMF each have a direct counterpart in this framework's methodology and findings.

NIST AI RMF Function	What This Framework Provides
GOVERN Establish policies and roles for	Five fidelity levels as a risk-tiering policy. Human-in-the-loop operating model at Level 5. Agent is research assistant, never t
MAP Identify and contextualize AI risks	Danger taxonomy: five failure modes documented across 757 test runs, ranked by severity. Each failure mode traced to spe
MEASURE Quantify risks with metrics and	Ground-truth test suites with verified answers. Scoring rubrics applied consistently across 3 use cases. Iterative retesting aft
MANAGE Act on risks and monitor over ti	Improvement Playbook: six-step cycle. Round-over-round regression testing. Model comparison across 6 configurations.

Trust but Verify as an Operating Model

At every level where consequences are material (Level 3 and above), this framework treats AI output as a draft, never a decision. The agent surfaces evidence, timelines, and discrepancies. A human reviewer evaluates that output against source records before any action is taken. This is not a guardrail bolted on after deployment - it is the design premise of every agent tested in this report.

This framework is complementary to, not a replacement for, an organization's broader responsible AI governance. The fidelity levels, danger taxonomy, and improvement playbook are designed to slot into an existing NIST AI RMF process - providing the MEASURE and MAP functions with empirical data while reinforcing the GOVERN and MANAGE functions with concrete operating procedures.

Conclusion

Not all AI use cases require the same investment. Levels 1 and 2 work with existing Copilot and Copilot Studio licenses, plus SharePoint document libraries or knowledge base uploads. From Level 3 onward, structured data connections via MCP dramatically improve fidelity - whether through Dataverse's built-in MCP connector, a custom MCP server, or both. The higher the level, the more you also need models with strong multi-step reasoning (GPT-4.1 in our testing), iterative testing, and human review workflows.

Use Case 3 added a critical dimension: **retrieval infrastructure matters as much as document quality.** The same enriched documents scored 39% through SharePoint search and 91% through knowledge base upload or Azure AI Search. Organizations should test their retrieval path before investing in model upgrades or custom engineering.

Copilot Studio agents are included with existing Microsoft 365 and Power Platform licenses. Foundry and pro-code agents use consumption-based Azure OpenAI pricing. The licensing cost scales with the level of investment, not the number of agents.

The Bottom Line

The agent is a research assistant, never the decision-maker. At Level 5, where legal outcomes depend on response fidelity, trust but verify is not a suggestion - it is the only responsible operating model.

Next Steps

1. **Start simple**
   Deploy a Copilot Studio agent with SharePoint grounding or knowledge base upload on a real document library. Measure it against known answers before customizing anything.
2. **Build your ground truth**
   Build a 10-prompt test suite with verified answers for your use case. Independently verify the existing baseline - do not trust self-reported accuracy. You cannot improve what you cannot measure.
3. **Test the retrieval path**
   Before investing in model upgrades or custom engineering, verify that your retrieval infrastructure surfaces the right documents. If it does not, no model will produce the right answer.
4. **Scale when ready**
   If your fidelity target exceeds what document agents deliver, connect structured data via MCP and follow the Improvement Playbook.
5. **Get started**
   Contact your Microsoft account team to scope a pilot.

Based on 757 test runs across 3 government use cases, 34 agent configurations, 6 models, 45 test prompts, and multiple rounds of iterative improvement.

Appendix: Meet Your Agents

Every score in this report comes from one of these agents. Here is who was tested and how they are built.

Use Case 1: Legal Case Analysis (17 Agents)

Agent	Data Source	Model	Final Score
Case Analyst Agent (pro-code)	MCP / SQL	GPT-4.1	10/10
Copilot Studio MCP/Com	MCP / SQL	GPT-4.1	10/10
Copilot Studio SP/PDF/Com	SharePoint PDFs	GPT-4.1	10/10
Copilot Studio KB/DOCX/Com	Uploaded DOCXs	GPT-4.1	10/10
Copilot Studio DV/Com (Sonnet)	Dataverse MCP	Sonnet 4.6	10/10
Copilot Studio DV/Com (5 Reasoning)	Dataverse MCP	GPT-5 Reasoning	10/10
Copilot Studio SP/DOCX/Com	SharePoint DOCXs	GPT-4.1	9/10
Copilot Studio MCP/GCC	MCP / SQL	GPT-4o	9/10
Copilot Studio SP/PDF/GCC	SharePoint PDFs	GPT-4o	9/10
Copilot Studio KB/PDF/GCC	Uploaded PDFs	GPT-4o	9/10
CS Custom Connector/GCC	APIM / SQL	GPT-4o	8.5/10
Copilot Studio KB/PDF/Com	Uploaded PDFs	GPT-4.1	8/10
Copilot Studio KB/DOCX/GCC	Uploaded DOCXs	GPT-4o	8/10
Copilot Studio SP/DOCX/GCC	SharePoint DOCXs	GPT-4o	7/10
Copilot Studio DV/Com (4.1)	Dataverse MCP	GPT-4.1	6/10
Copilot Studio DV/Com (5 Auto)	Dataverse MCP	GPT-5 Auto	4/10
Copilot Studio DV/GCC	Dataverse MCP	GPT-4o	3.2/10

Ranked by final score. All agents retested with cross-referenced documents. PDF outperformed DOCX in 3 of 4 matchups; KB/DOCX/Com was the exception (10/10). Dataverse MCP tested across 5 models: Sonnet 4.6 achieved 10/10 (perfect); GPT-5 Reasoning 10/10; GPT-4o/GCC reached 3.2/10 after 6 rounds of optimization.

Use Case 2: Investigative Analytics (8 Agents)

Agent	Platform	Model	Final Score
Copilot Studio MCP/Com	Copilot Studio	GPT-4.1	10/10
Investigative Agent	OpenAI SDK	GPT-4.1	10/10
Triage Agent	Semantic Kernel	GPT-4.1	10/10
Foundry Agent	Foundry Agent Service	GPT-4.1	9/10
Copilot Studio SP/PDF/GCC	Copilot Studio	GPT-4o	9/10
Copilot Studio SP/PDF/Com	Copilot Studio	GPT-4.1	8/10
Copilot Studio MCP/GCC	Copilot Studio	GPT-4o	4/10

M365 Copilot MCP/Com	M365 Platform	Platform-assigned	2/10

Ranked by final score. Case Analyst Agent tested on UC1 only.

Use Case 3: Policy Compliance (12 Agents)

Agent	Retrieval Method	Model	Accuracy
GCC KB xRef	KB Upload	GPT-4o	91% (21/23)
Azure AI Search Enriched	AI Search	GPT-4.1	91% (21/23)
CPS AI Search Enriched	AI Search + PA	GPT-4o	91% (21/23)
Azure AI Search Baseline	AI Search	GPT-4.1	70% (16/23)
GCC SP XRef + Meta	SharePoint	GPT-4o	61% (14/23)
GCC SP XRef	SharePoint	GPT-4o	52% (12/23)
DHHS GCC KB Baseline	KB Upload	GPT-4o	57% (13/23)*
GCC SP Baseline	SharePoint	GPT-4o	39% (9/23)
COM SP Baseline	SharePoint	GPT-4.1	39% (9/23)
GCC Web	Web KB	GPT-4o	39% (9/23)
GCC KB Baseline	KB Upload	GPT-4o	39% (9/23)
GCC SP XRef PDF	SharePoint	GPT-4o	39% (9/23)

** DHHS baseline adjusted from customer-reported 70% (16/23) to 57% (13/23) after independent grading review. Ranked by accuracy. PA = Power Automate. COM SP Baseline (GPT-4.1) = 39% confirms the model is not the bottleneck when retrieval fails.*

Appendix: Test Prompts

Every agent was tested against the same prompts per use case to ensure consistent, comparable scoring. Prompts progress from simple retrieval to multi-step investigative reasoning.

Use Case 1: DSS Legal (Document Agents)

#	Prompt	Tests
1	What was the ER admission time for Jaylen Webb, and who was the nurse?	Simple factual retrieval
2	What did Marcus Webb say to hospital staff vs. law enforcement about the night Jaylen was injured?	Cross-document statement comparison
3	Crystal Price told the court she was 'clean now' at the November 2023 hearing. What do the drug test results show?	Contradiction detection
4	Did the Sheriff's Office investigation find fractures in Jaylen Webb's skeletal survey?	Cross-document conflict resolution
5	What is the complete timeline of events for case 2024-DR-42-0892?	Full case narrative assembly
6	List all people involved in the Price TPR case and their roles.	Complete roster extraction
7	Compare Dena Holloway's initial hospital statement with her later statement to Lt. Odom. What changed?	Statement evolution tracking
8	What statements were made to law enforcement in case 2024-DR-42-0892?	Filtered statement retrieval
9	Which cases involve Termination of Parental Rights?	Multi-case aggregate query
10	What was the exact time gap between the thump Dena heard and when they took Jaylen to the hospital?	Precision time calculation

Use Case 2: Investigative Analytics (Data Agents)

#	Prompt	Tests
1	How many properties does GEENA LLC own, and what percentage are vacant?	Simple fact extraction
2	What happened to 4763 Griscom Street between when GEENA LLC bought it and today?	Cross-source synthesis
3	What was the assessment value of 4763 Griscom Street in 2017, and what is it now? What happened?	Numeric extraction + causation
4	Trace the complete ownership history of 4763 Griscom Street. How many times was it sold at sheriff sale?	Chain reconstruction
5	The transfer records show 4763 Griscom was sold for $146,000 in 2004. What was the fair market value at the time?	Contradiction detection
6	Who are the top 5 private property owners in Philadelphia by code violation count? Exclude government entities.	Citywide aggregate query
7	What financial institutions were involved in the mortgage history of 2400 Bryn Mawr Avenue?	Narrative context extraction

8	How many code violations at 4763 Griscom Street resulted in FAILED inspections, and what type of investigator handles them?	Filtering precision
9	Compare the vacancy rates and violation rates between zip codes 19104 and 19132. Which is worse?	Cross-area aggregate
10	GEENA LLC bought 2400 Bryn Mawr Ave for $230,000 at sheriff sale. The assessed value was $357,100. What percentage discount was that?	Multi-step arithmetic

Use Case 3: Policy Compliance (Document Agents)

#	Prompt	Tests
1	If a person applied for pregnancy coverage and reported 2 months later that pregnancy ended, will they get postpartum coverage?	Policy interpretation
1.1	Is the postpartum period applicable for PW in Emergency coverage?	Follow-up specificity
1.2	Can we assumptively approve Pregnancy women coverage for emergency services?	Policy boundary testing
2	If I transfer $10,000 to my brother, is there a penalty?	Asset transfer rules
2.1	How long is the penalty period?	Calculation methodology
3	Can we re-open an IS application within 30 days if the case was closed due to over resources?	Procedural rule lookup
4	What is the burial fund resource limit for a couple?	Simple threshold lookup
5	Tell me about COLA disregard	Open-ended policy explanation
6	Tell me about AVS	Broad topic retrieval
7	When should an AVS Geo search be requested for a new application?	Procedural specificity
8	When should an AVS FI search be requested for a new application?	Trigger condition identification
9	What process should be followed if an AVS FI search is not returned from the bank?	Fallback procedure
10	When is the cash value of Life insurance counted?	Conditional policy rule
10a	If a member applying for ABD reports Life insurance with FV $5000 and CSV $3000, will that resource be counted?	Scenario-based application
10b	Same question, per liberalized policy?	Policy framework selection
13	Tell me about Income trust	Topic retrieval
14	Can we accept electronic signatures on a 1282?	Specific form policy
15	Can a 1099R Form be used for verification of retirement income?	Document verification rules
19	Do we count Veterans benefits towards MAGI budgeting/PHC?	Cross-program income rules
20	Do we count child's SSA towards MAGI?	Income counting rules
21	Is a child's SSA counted for MAGI categories?	Rephrased consistency test
22	Is a child's SSA counted for MAGI categories? (same question, different user)	Cross-user consistency

UC3 prompts developed by state Medicaid policy experts. Questions 12 and 16 excluded from scoring (not tested across all agents). 23 of 25 questions scored. All 34 agents across 3 use cases were scored against these prompts.